

James Bona-Landry

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EDUCATION

University of Toronto, Toronto, Ontario, CA

- **Ph.D. in Mathematics**
• Research interests: stochastic analysis, KPZ universality, integrable probability
• Advisor: [Prof. Jeremy Quastel](#)
• Expected Graduation Date: Aug 2028
- **M.Sc. in Mathematics**
• Thesis: [Stationarity of Brownian Motion for the Open KPZ Equation](#)
• Advisor: [Prof. Jeremy Quastel](#)
• Coursework spanning stochastic analysis, PDEs, algebraic complexity theory, and deep learning
- **B.Sc. (Hons) with High Distinction, Mathematics Specialist**
• Cumulative GPA: 3.92 / 4.0
• Completed graduate coursework requirements of Ph.D. program in addition to undergraduate curriculum

Sep 2025 – Present

Sep 2024 – Nov 2025

Sep 2020 – Jun 2024

EXPERIENCE

- **Mathematics Researcher**, University of Toronto
• Studying the large-scale asymptotic behaviour of random growth models and interacting particle systems, with a focus on invariant measures, scaling limits, and universality phenomena
- **Teaching Assistant**, University of Toronto
• Courses: [MAT1128H](#) (Stochastic Differential Equations), [MAT344H](#) (Combinatorics), [MAT292H](#) (ODEs for Engineers), [MAT244H](#) (ODEs), [MAT236H](#) (Vector Calculus II), [MAT235H](#) (Vector Calculus I), [MAT136H](#) (Calculus II), [MAT135H](#) (Calculus I), [PUMP I](#)
• Duties include running tutorials, hosting office hours, grading coursework, and invigilating exams
- **Conference Organizer / Chair**, University of Toronto
• Bird's Eye Conference (2026): chair, probability section
• Canadian Undergraduate Mathematics Conference (2023): senior organizer, ~ 200 attendees

Apr 2024 – Present

Sep 2023 – Present

Jun 2023, Feb 2026

PROJECTS & PUBLICATIONS

- **Invariant Measures for the Open KPZ Equation: the Gaussian Case**, *Preprint*, James Bona-Landry ([arXiv](#), [PDF](#))
• Submitted for publication to *Mathematical Physics, Analysis and Geometry*
• Showed that for a certain class of boundary conditions, the open KPZ equation has (modulo a height shift) a Gaussian invariant measure given by Brownian motion with constant drift
- **FeatureFusion**, *Project*, James Bona-Landry and Murdock Aubry ([GitHub](#), [PDF](#))
• Introduced, analyzed, and evaluated a novel method for merging diffusion models trained on different tasks based on the correlations between their internal feature representations
• Rigorously proved formula for approximate sampling distribution of merged model

Apr 2026

Apr 2025

TECHNICAL SKILLS

- **Applied Mathematics:** Probability, stochastic calculus, (stochastic) PDEs, numerical & statistical methods, deep learning
- **Programming:** Python (PyTorch, NumPy), HTML/CSS, $\text{\LaTeX}$

AWARDS & SCHOLARSHIPS

- **Mathematics Department Research Award**, University of Toronto
Funded a summer research project on integrable statistical mechanics under the supervision of [Prof. Jeremy Quastel](#) ([PDF of Write-Up](#))
- **NSERC Undergraduate Student Research Award**, University of Toronto
Funded a summer research project on spin chains under the supervision of [Prof. Marco Gualtieri](#) ([PDF of Write-Up](#))

Apr 2023 – Aug 2023

Apr 2022 – Aug 2022

[Last updated 2026-09-24]